



Timescales of magmatic-hydrothermal activity at the giant San Rafael tin deposit (Peru)

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ABSTRACT

Recent research suggests that protracted magmatic activity may play a critical role in forming large metal endowments in magmatic-hydrothermal deposits. Here, we test whether this concept applies to tin systems associated with peraluminous (S-type) granites, focusing on the intrusive complex at San Rafael (Peru) that hosts the world's largest and highest-grade primary Sn deposit (>1.5 Mt of Sn at 3.7 %). We obtained a comprehensive ($n = 187$) set of high-precision U-Pb zircon dates from granites and mafic rocks that both pre- and post-date the cassiterite Sn ore according to crosscutting relationships. Individual granites exhibit large date dispersions (up to 1.7 Myr), which we evaluate in the context of magmatic and hydrothermal zircon origins, possibility of unmitigated Pb loss, and cryptic inclusion of older zircon cores. By integrating zircon compositions with field observations, we reconstruct 200–400 kyr of magmatic evolution in the crust – from 24.402 ± 0.019 Ma to final emplacement at shallow levels between 24.177 ± 0.004 Ma and 24.027 ± 0.028 Ma. Younger ages found in one sample of the ore-stage cassiterite (23.80 ± 0.25 Ma) and all samples of hydrothermal adularia (~ 22.72 – 22.43 Ma) suggest that the mineralizing fluids were most likely not related to the cooling of the exposed intrusive complex, but instead to one or more younger magma pulses outgassing at depth. The duration of the magmatic build-up prior to Sn mineralization at San Rafael appears significantly shorter than in the largest porphyry copper systems, although we recognize that longer timescales may be resolvable in the unexposed portions of the system. Finally, we compared the metal tonnage and computed volumes of metal-sourcing melt at San Rafael (1.5 Mt of Sn and 77 km^3 of melt) with the largest porphyry copper systems in magmatic arcs (112 Mt of Cu and $>1000 \text{ km}^3$ of melt) to propose that more limited and episodic melt supply, alongside with restricted water availability in (post-)collisional S-type systems may limit the maximum endowments of Sn systems.

1. Introduction

Determining the durations of magmatic-hydrothermal processes and the exact timing of ore-forming events is essential for the understanding of mineral systems and calibrating models of emplacement, evolution, and degassing of magmatic bodies (e.g., Annen, 2009; Chelle-Michou et al., 2017a). Modern high-precision zircon U-Pb geochronology has greatly advanced our knowledge of calc-alkaline, arc-related magmatism associated with porphyry copper systems, revealing that mineralization events in ore districts are brief compared to the multimillion-year

build-up of source magma bodies (e.g., Large et al., 2020; Virmond et al., 2024). In contrast to porphyry copper systems, primary tin (Sn) mineral systems are most often associated with relatively reduced (low $f\text{O}_2$), peraluminous S-type granitic magmatism (Černý et al., 2005; Lehmann, 2021). Tin in reduced magmas (Sn^{2+}) behaves as an incompatible element, effectively partitioning into the magmatic-hydrothermal fluid upon cooling and degassing of a magma body (e.g., Patterson et al., 1981; Heinrich, 1990; Keppler and Audétat, 2025). Tin precipitates from the fluid through cooling, depressurization, oxidation upon mixing with meteoric waters, or fluid-rock interaction, primarily as cassiterite (SnO_2 ;

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Sn^{4+} in veins, lodes and greisens (Korges et al., 2018; Harlaux et al., 2021b; Torró et al., 2024). While this model explains the genesis of Sn deposits in S-type systems, the timescales over which mineralizing S-type magmas are generated at depth and assembled in the upper crust remain largely unknown. This represents a significant knowledge gap, as timescales have been shown to directly influence metal endowments in other magmatic-hydrothermal systems (Large et al., 2020; Virmond et al., 2024).

Here, we investigate the timescales of magmatic-hydrothermal processes that formed the San Rafael Sn(-Cu) deposit using high-precision zircon U-Pb isotope analyses by isotope dilution thermal ionization mass spectrometry (ID-TIMS). San Rafael represents one of the best studied and largest tin deposits globally, with past production, reserves, and resources combined exceeding 1.5 Mt of Sn at an average mined ore grade of 3.7 % Sn (e.g., Kontak and Clark, 2002; Harlaux et al., 2023). We apply zircon geochronology and complementary compositional analyses (i.e., petrochronology) to a suite of magmatic units from San Rafael, which provides us with valuable insights into petrologic and metallogenic processes from magma generation in the lower to middle crust to cooling and outgassing of the magma reservoir in the upper crust. Complementary to zircon petrochronology, we carried out in situ U-Pb cassiterite dating to independently constrain the absolute age(s) of the Sn mineralization and further inform our interpretations. Together,

these results allow for a comprehensive inquiry into the magmatic-hydrothermal timescales in a Sn-ore-forming, S-type system at high precision.

2. Geological background

The San Rafael Sn(-Cu) deposit is located in southern Peru, within the Eastern Cordilleran Macusani Structural Zone (Perez et al., 2016), also known as the pre-Cordillera de Carabaya (Fig. 1A; Clark et al., 1983, 1990; Sandeman et al., 1995). It represents one of the northernmost deposits in the Central Andean Tin Belt, which contains hundreds of Sn-W ± Ag-Zn-Pb-Cu deposits and occurrences of either late Triassic or late Oligocene–early Miocene age (e.g., Lehmann et al., 1990; Gemmrich et al., 2021; Torró et al., 2024). Tin mineralization at San Rafael is associated with an Oligocene intrusive complex (~24 Ma; e.g., Harlaux et al., 2021a) composed of a suite of peraluminous granites and subordinate mafic dykes (Sandeman et al., 1997). The mineralization is hosted in NNW-trending, cassiterite-bearing lodes that extend vertically for at least 1.6 km, from within the granite into the overlying metasedimentary cover (Fig. 1B and C; e.g., Kontak and Clark, 2002; Gialli et al., 2017; Harlaux et al., 2023; Minsur, 2023).

The intrusive complex was emplaced at depths of 2–3 km (based on fluid inclusion homogenization pressures of Kontak and Clark, 2002)

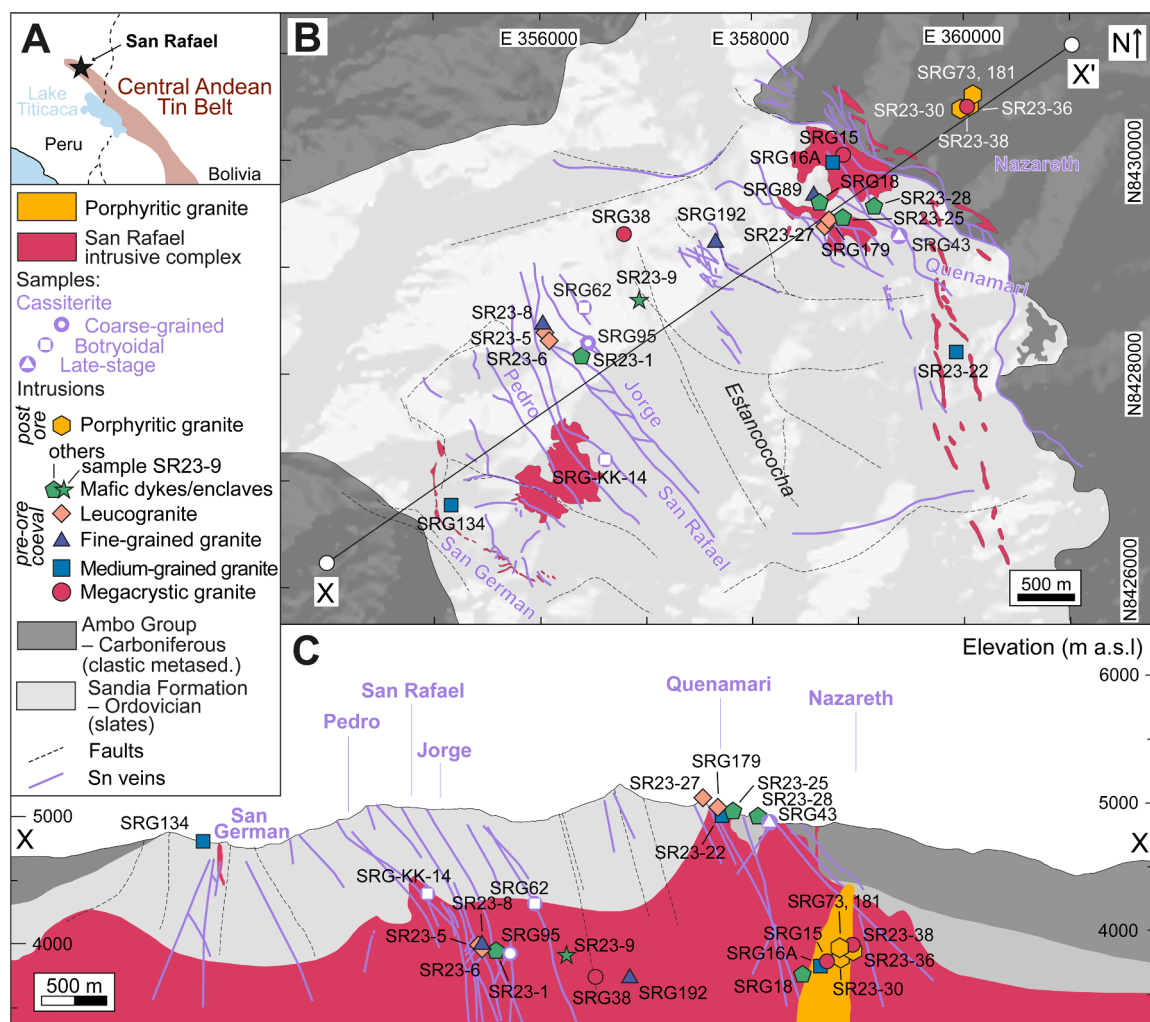


Fig. 1. (A) Location of the San Rafael Sn(-Cu) deposit within the Central Andean Tin Belt. (B) Geological map and (C) cross section (X–X') showing the Oligocene intrusive complex, major faults, and distribution of Sn vein systems (modified after Harlaux et al., 2023; updated with data from Minsur, 2023). The megacrystic granite constitutes the bulk of the intrusive complex, whereas other granite units, except the porphyritic granite, are not resolved at this scale. Sample locations in (C) are projected onto the cross-section.

into Paleozoic metasedimentary rocks of the Sandia Formation and Ambo Group (Fig. 1C; Laubacher, 1978). This magmatism was formed in a post-collisional setting dominated by the retreat of the Nazca and Farallon slabs (~30–22 Ma), which promoted asthenospheric upwelling, decompression melting of the upper mantle, and elevated heat flux in the Macusani Structural Zone (e.g., Carlier et al., 2005, Mamani et al., 2010). The granitic magmas at San Rafael likely originated through partial melting of overthickened mid- to lower-crustal metasedimentary sequences, which was locally intensified by injections of mantle-derived magmas (Harlaux et al., 2021a).

3. Intrusive suite and cassiterite mineralization at San Rafael

Five granitic units are recognized at San Rafael based on macroscopic texture: 1) megacrystic granite, 2) medium-grained granite, 3) fine-grained granite, 4) leucogranite, and 5) porphyritic granite (Figs. 2

and S1). Descriptions of these units are provided in the Supplementary Material, while detailed petrographic data are provided in Kontak and Clark (2002) and Harlaux et al. (2021a). All granites share a peraluminous character with a magmatic mineral assemblage of K-feldspar, plagioclase, quartz, biotite, cordierite $\pm$ tourmaline, and accessory zircon, apatite, ilmenite, and monazite. The megacrystic granite comprises over 90 % of the exposed complex, whereas the other granites occur in centimeter- to meter-scale intervals in outcrops and drill cores (Figs. 1 and 2). The megacrystic, medium-grained, and fine-grained granites (Fig. 2A–C) exhibit mutual co-magmatic enclaves with sharp to gradual contacts implying roughly coeval crystallization at the exposure level. The leucogranite (Fig. 2D) forms dykes that crosscut the megacrystic granite (Harlaux et al., 2021a). A later magmatic phase was identified in the Nazareth area (Fig. 1) at >500 m depth as a 100–400 m thick dyke of porphyritic granite that crosscuts the mineralized megacrystic granite (Fig. 2H–I; Harlaux et al., 2021a). Mafic units (also

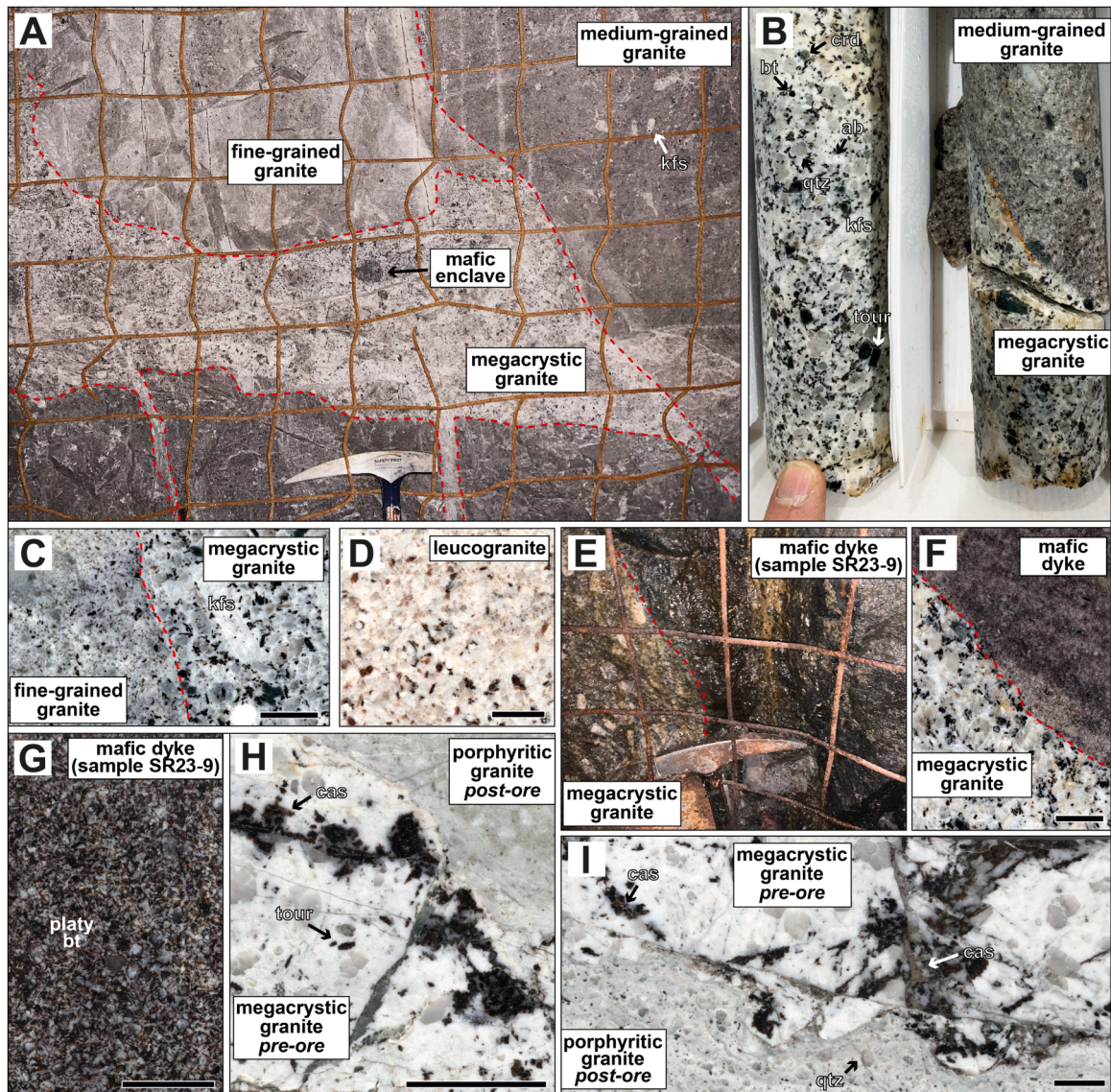


Fig. 2. Textures of San Rafael intrusions. (A) Outcrop-scale view showing the relationships between the megacrystic, medium-grained, and fine-grained granites. Contacts between the megacrystic, and the medium-grained (B) and fine-grained granites (C). (D) Macroscopic texture of the leucogranite, distinguished from the fine-grained granite by its larger phenocrysts and lower biotite abundance. (E) Mafic dyke (sample SR23–9) crosscutting the megacrystic granite. (F) Contact between a mafic dyke and megacrystic granite observed in drill core. (G) Macroscopic texture of the mafic dyke shown in E, characterized by abundant biotite. (H–I) Drill core intervals from the Nazareth area showing the barren porphyritic granite crosscutting the ore-bearing megacrystic granite (mineralized in hydrothermal stage II; Harlaux et al., 2023). bt – biotite, cas – cassiterite (brown), crd – cordierite, kfs – K-feldspar, plg – plagioclase, tour – tourmaline (black), qtz – quartz. Scale bars are 1 cm.

referred to as lamprophyres) occur locally as enclave swarms and dismembered dykes within the megacrystic granite, or as later dykes that intrude into it (Fig. 2E–G).

The hydrothermal evolution of San Rafael, described by Kontak and Clark (2002) and updated by Harlaux et al. (2023; see Fig. 3), includes five stages: (0) early pervasive alkali metasomatism, (I) ore-barren greisenization and quartz-tourmaline veining/brecciation, (II) main cassiterite ore mineralization, (III) minor polymetallic sulfide and cassiterite mineralization, and (IV) quartz-carbonate veining (Fig. 3). Tin ore at San Rafael occurs as coarse-grained and botryoidal cassiterite (also known as “wood tin”), hosted within quartz-chlorite ± tourmaline-bearing veins, lodes, and hydrothermal breccias. These cassiterite-bearing structures crosscut the megacrystic, medium-grained, and fine-grained granites, as well as mafic dykes, indicating that these units were fully crystallized prior to Sn ore deposition. The relationship between ore-stage cassiterite and leucogranite remains unclear, as they have not been observed in direct contact. The porphyritic granite crosscuts the main-stage cassiterite mineralization, thereby bracketing the precipitation of Sn ore (Fig. 2H–I). The porphyritic granite is itself crosscut by late-stage polymetallic veins, assigned to hydrothermal stages III and IV of Harlaux et al. (2023; Fig. 3), which are associated with adularia and minor amounts of fine-grained, sub-economic cassiterite. All magmatic units exhibit variable hydrothermal

alteration (e.g., tourmaline, adularia, chlorite etc.), with the strongest overprint observed in the late porphyritic granite (Table S1; Harlaux et al., 2021a).

4. Samples and methods

Eighteen samples of intrusive rocks and five of cassiterite ore were collected from outcrops, underground galleries, and drill cores at San Rafael (Fig. 1B–C and Table S1). Zircon crystals were extracted from the rock samples by heavy liquid (methylene iodide) gravity separation, annealed at 900 °C for 48 h, and mounted in epoxy. Internal zircon textures were characterized by cathodoluminescence (CL) imaging on a JEOL JSM-6390 scanning electron microscope (SEM). In situ U-Pb dating and trace-element analyses of zircon cores and rims were performed with a S155-LR ASI Resolution Excimer laser ablation (LA) system coupled to a Thermo Element-XR sector field single collector inductively coupled plasma mass spectrometer (ICPMS) at ETH Zurich. CL imaging and in situ U-Pb screening prior to zircon dissolution for high-precision dating is particularly important in peraluminous systems, where >90 % of zircon crystals in a sample may be inherited from older lithologies (Couzinié et al., 2021). After screening, 187 whole zircon crystals from all magmatic units at San Rafael were dissolved and analyzed for high-precision U-Pb geochronology using isotope dilution

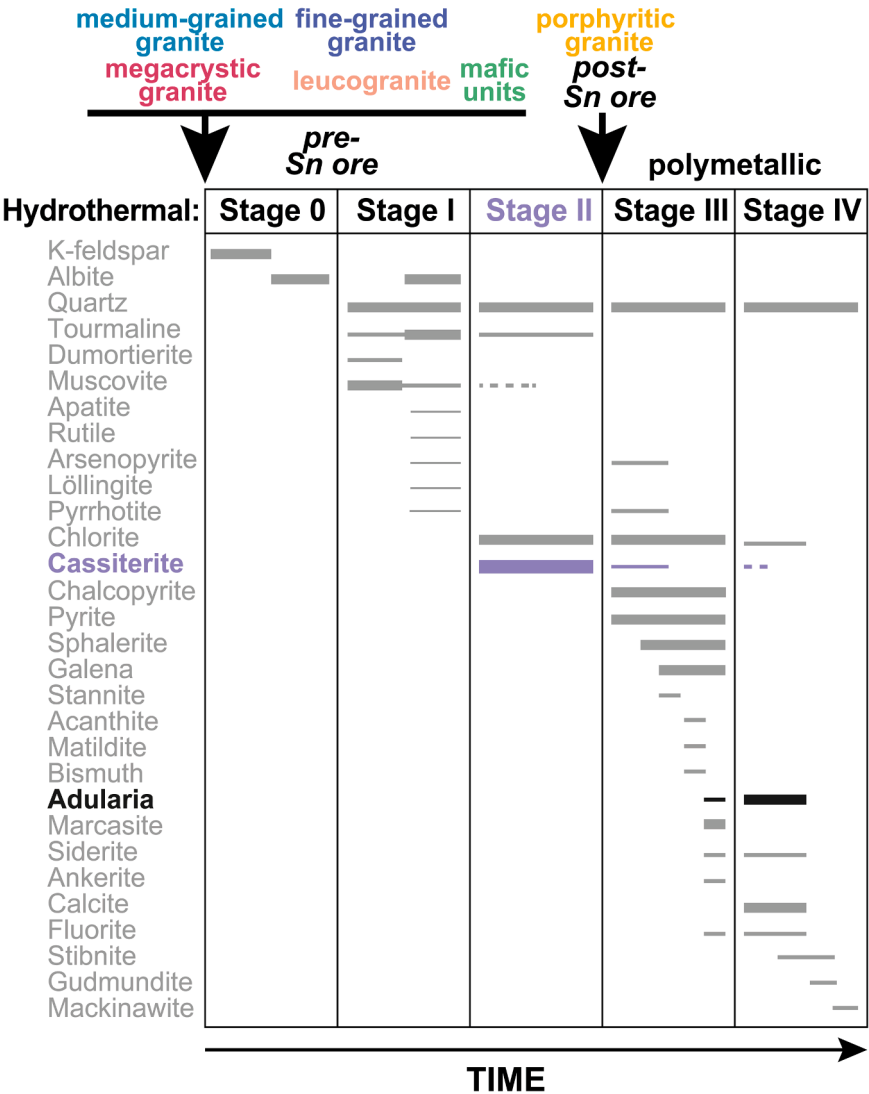


Fig. 3. Sequence of magmatic units and hydrothermal stages at San Rafael (modified after Harlaux et al., 2023). Line thicknesses represent relative mineral abundance.

thermal ionization mass spectrometry (ID-TIMS) at ETH Zurich between October 2022 and July 2024. Prior to dissolution, crystals were chemically abraded in HF at 190–210 °C for 10–14 h to remove radiation-damaged domains that could exhibit Pb loss (Mattinson, 2005; McKanna et al., 2024). High-precision Pb and U (as UO_2) isotope ratios were measured on a Thermo Triton Plus TIMS equipped with Faraday cups connected to $10^{13} \Omega$ amplifiers and an axial secondary electron multiplier (von Quadt et al., 2016; Wotzlaw et al., 2017). The resulting $^{206}\text{Pb}/^{238}\text{U}$ dates were corrected for initial $^{230}\text{Th}/^{238}\text{U}$ disequilibrium

and are reported with 2σ analytical uncertainties, excluding systematic uncertainties such as those related to spike calibration or U decay constant.

Cassiterite samples – including coarse-grained and botryoidal ore-stage varieties (stage II; Fig. 3) as well as late-stage fine-grained cassiterite (stage IV) – were analyzed in situ for U-Pb isotopes using a RESolution S-155 193-nm ArF excimer LA system coupled with an Thermo Element XR sector-field ICPMS at ETH Zurich following Bermejo et al. (2025). Full analytical procedures are provided in the Supplementary

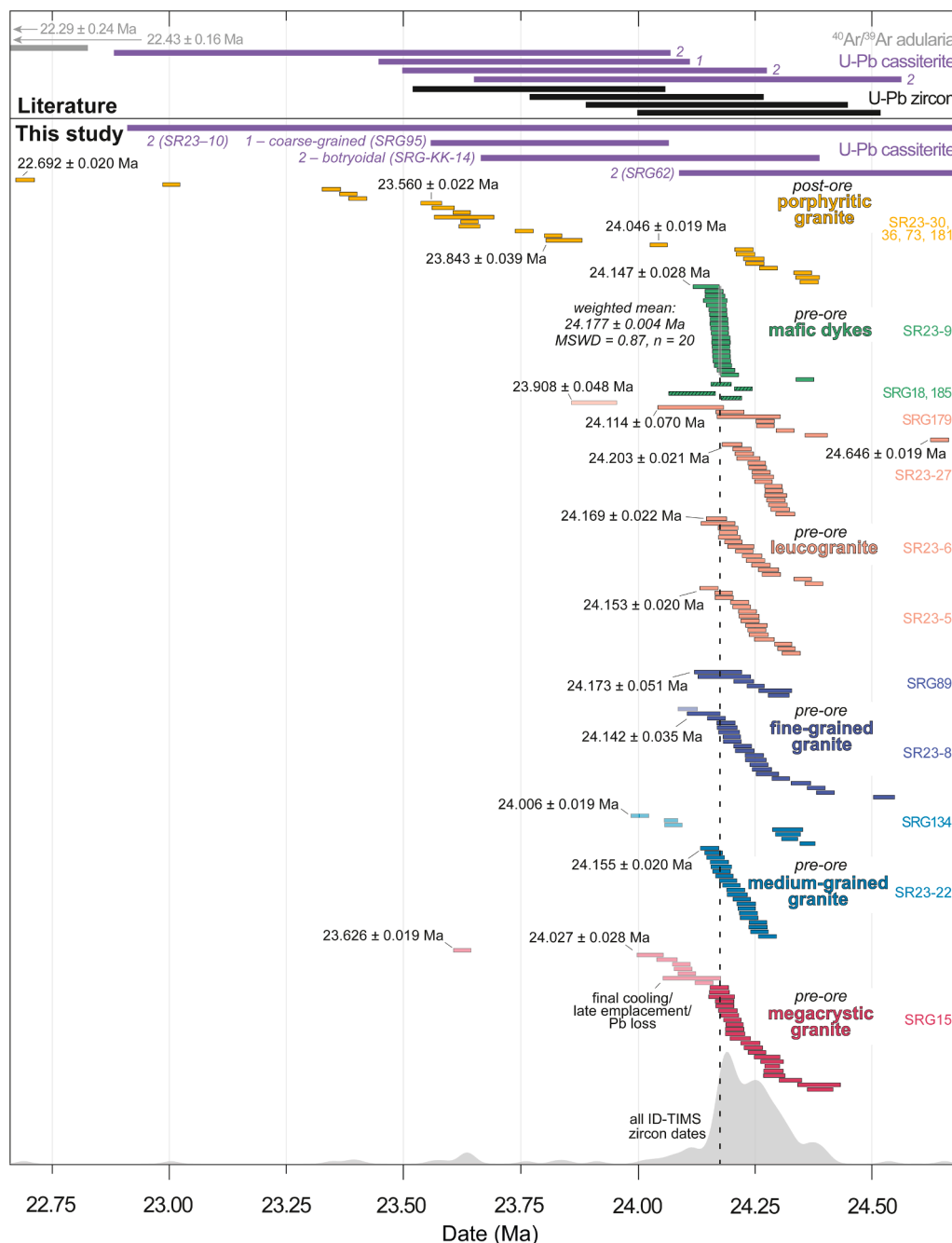


Fig. 4. Geochronology of the Sn(-Cu) magmatic-hydrothermal system at San Rafael. Smaller colored bars represent single-crystal $^{206}\text{Pb}/^{238}\text{U}$ zircon dates obtained by ID-TIMS, plotted with their respective 2σ analytical uncertainties. Dates are age-ranked by sample, listed from bottom to top as in Table S2. For the porphyritic granite, zircon dates from four samples from a single drillhole are plotted together due to sample proximity. Wider purple bars show lower intercept cassiterite dates with propagated external uncertainty (1.8 % from Carr et al., 2023), from this study and from Harlaux et al. (2023). Black bars show in situ zircon weighted mean $^{206}\text{Pb}/^{238}\text{U}$ ages from Harlaux et al. (2021a). Also shown are $^{40}\text{Ar}/^{39}\text{Ar}$ plateau ages for hydrothermal adularia associated with late-stage polymetallic veins (hydrothermal stages III and IV of Harlaux et al., 2023), recalculated relative to Fish Canyon Tuff sanidine (Kuiper et al., 2008). MSWD – mean square of weighted deviates.

Material.

5. Results

5.1. Zircon CL textures

The Oligocene zircon populations (as opposed to older, inherited zircon) from the pre-ore granites occur as bipyramidal crystals ranging from 30 to 400 μm in length (c axis). They exhibit various CL textures, with no systematic correlation to the host unit (Fig. S2). Most exhibit concentric oscillatory, irregular or patchy zoning, either throughout the entire crystal or restricted to rims. Fewer crystals display dark CL cores with brighter intergrowths, and faint zoning. Bright cores, both Oligocene and older, overgrown by Oligocene dark oscillatory rims are observed across all granite units. Zircon crystals from the post-ore porphyritic granite are, on average, darker in CL tone and resemble the dark oscillatory-zoned rims found in other granite units. Among the mafic dykes, five samples (SR23-1, 25, 28 and SRG18, 185) contain crystals that are indistinguishable from those from the granites in terms

of size and internal texture. In contrast, sample SR23-9 hosts a distinct zircon population (population I in Fig. S2), composed predominantly of prismatic crystals (>200 μm long) with weak zoning and no evidence for inheritance.

5.2. U-Pb zircon and cassiterite geochronology

High-precision $^{206}\text{Pb}/^{238}\text{U}$ dates of Oligocene zircon (all from concordant analyses) range between 24.646 ± 0.019 Ma and 22.692 ± 0.020 Ma (Fig. 4 and Table S2). Two outliers older than 24.5 Ma from the fine-grained granite and leucogranite units, which may either represent inheritance or mixed dates, are not further discussed. Within individual granite samples, the zircon dates record up to ~ 250 kyr of dispersion, with the notable exception of those from the porphyritic granite that spread over 1.7 Myr. Most dates, irrespective of the host unit or sample, fall between 24.402 ± 0.019 Ma and 24.006 ± 0.019 Ma. Within this group, four dates from two mafic dyke samples (SRG18 and SRG185) scatter between 24.227 ± 0.019 Ma and 24.117 ± 0.050 Ma. They are characterized by low Th/U (<0.12), indistinguishable from

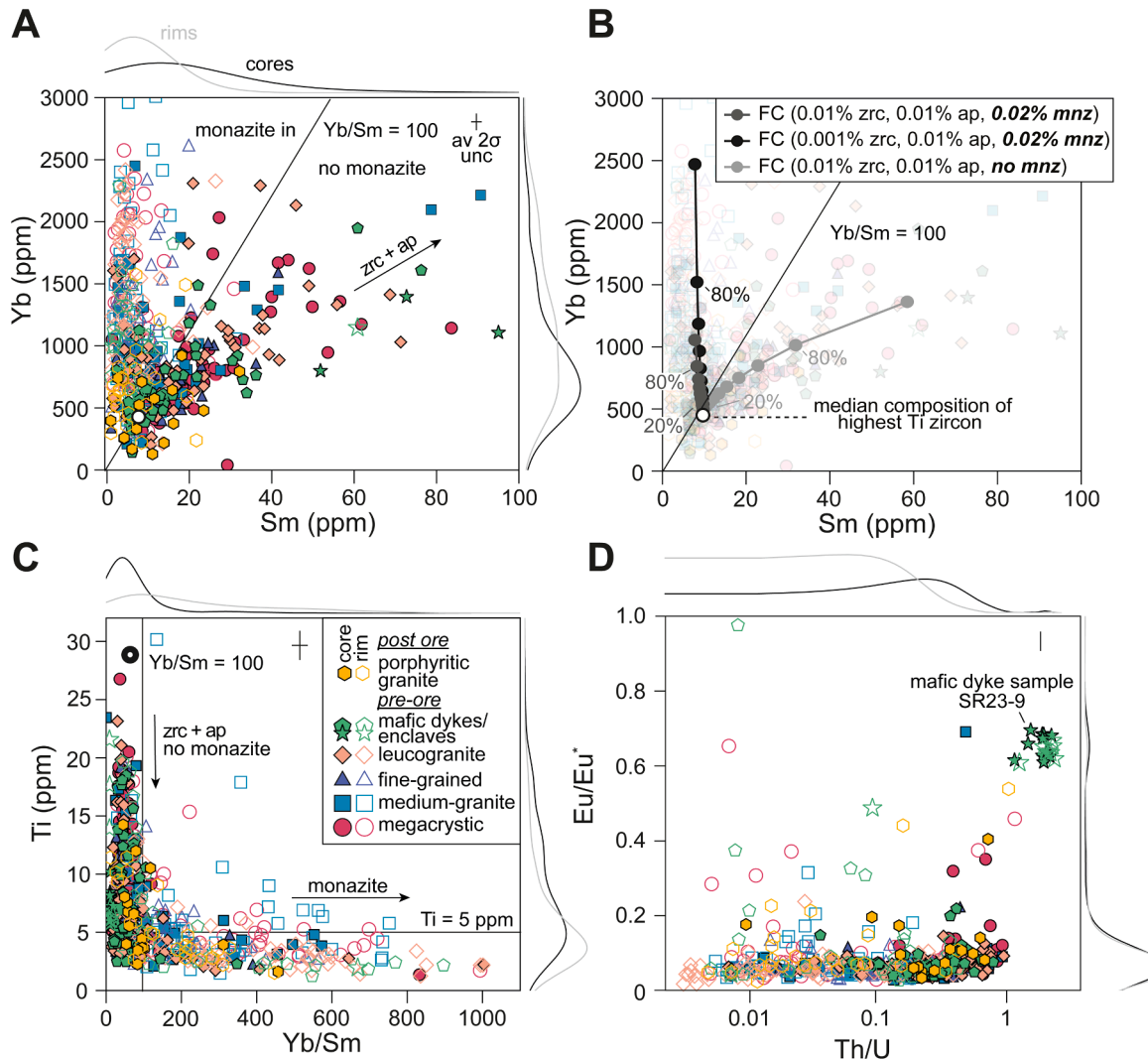


Fig. 5. Core-rim geochemical signatures of Oligocene zircon from San Rafael. (A) High and low Yb/Sm trends recorded by zircon cores and rims across different units. (B) Modelled Yb and Sm contents in zircon as a function of varying proportions of accessory minerals zircon, apatite, and monazite in the crystallizing assemblage (arrows in A and B). The starting composition corresponds to that of zircon with the highest Ti content, assumed to have crystallized from the least evolved melt (see Supplementary Material for modelling details). (C) Evolution of core and rim Yb/Sm ratios as a function of zircon Ti content used as a proxy for magma temperature. (D) Low Th/U-low Eu/Eu* zircon from the granites and most mafic dykes, and high Th/U-high Eu/Eu* zircon from the mafic dyke sample SR23-9. Kernel density estimates (KDEs) for each panel are shown for all cores (black line) and rims (grey line), irrespective of unit. ap – apatite, mnz – monazite, zrc – zircon; av 2σ unc – average 2σ uncertainty.

zircon from the granites. In contrast, the prismatic zircon crystals from the mafic dyke sample SR23–9 are characterized by high Th/U (>1.66) and all but one ($n = 20$, excluding one date at 24.358 ± 0.019 Ma) return indistinguishable dates allowing to calculate a weighted mean date at 24.177 ± 0.004 Ma (Fig. 4). Younger dates, ranging from 23.908 ± 0.048 Ma to 22.692 ± 0.020 Ma, are recorded by one crystal from the megacrystic granite and the leucogranite each, and fourteen crystals in the porphyritic granite.

A sample of coarse-grained cassiterite (SRG95) from a quartz-chlorite vein returned a lower intercept date at 23.80 ± 0.25 Ma (MSWD = 1.20, $n = 117$; see Fig. S4A and Table S3). Three botryoidal cassiterite samples from a hydrothermal breccia body yielded overlapping lower intercept U–Pb dates of 24.43 ± 0.32 Ma (MSWD = 0.36, $n = 119$, sample SRG62; Fig. S43B), 24.02 ± 0.34 Ma (MSWD = 0.48, $n = 106$, sample SRG-KK-14; Fig. S4C) and 24.08 ± 1.17 Ma (MSWD = 4.3, $n = 157$, SR23–10; Fig. S5). Analyses of the late-stage cassiterite (SRG43) showed high common Pb contents, preventing a meaningful age determination.

5.3. Zircon geochemistry

In situ trace element analyses of Oligocene zircon from San Rafael reveal diverging compositional trends recorded by cores and rims (Fig. 5A and Table S3). Most core compositions show a positive

correlation between MREE and HREE, with Sm contents ranging from 15 to 100 ppm and Yb from 750 to 2200 ppm ($\text{Yb}/\text{Sm} < 100$). In contrast, nearly all rims display highly variable Yb contents (100–3000 ppm) but consistently low Sm contents (< 15 ppm), resulting in elevated Yb/Sm ratios (100–1000). Titanium contents in cores span a broad range from 2 to 27 ppm, whereas rims typically display a narrower range from 2 to 7 ppm (Fig. 5C).

Most granitic zircon analyses return low Th/U ratios between 0.04–0.30 and Eu/Eu^* below 0.15, while zircon crystals from the mafic dyke sample SR23–9 show higher Th/U (> 1.9) and Eu/Eu^* between 0.6–0.7 (Fig. 4D). Some cores and rims from various units fall between these two populations in terms of Th/U and Eu/Eu^* . In the dominant, low Th/U–low Eu/Eu^* population, cores generally have higher Th/U (> 0.15) than the rims ($\text{Th}/\text{U} < 0.15$). A notable feature of zircon from both the granites and mafic units at San Rafael is their high U content, with most analyses exceeding 1000 ppm U and some reaching up to 19,000 ppm (Fig. S5). Importantly, there are no systematic differences in zircon composition between the granite units or hand-samples (Figs. S6 and S7 for unit- and sample-based comparisons).

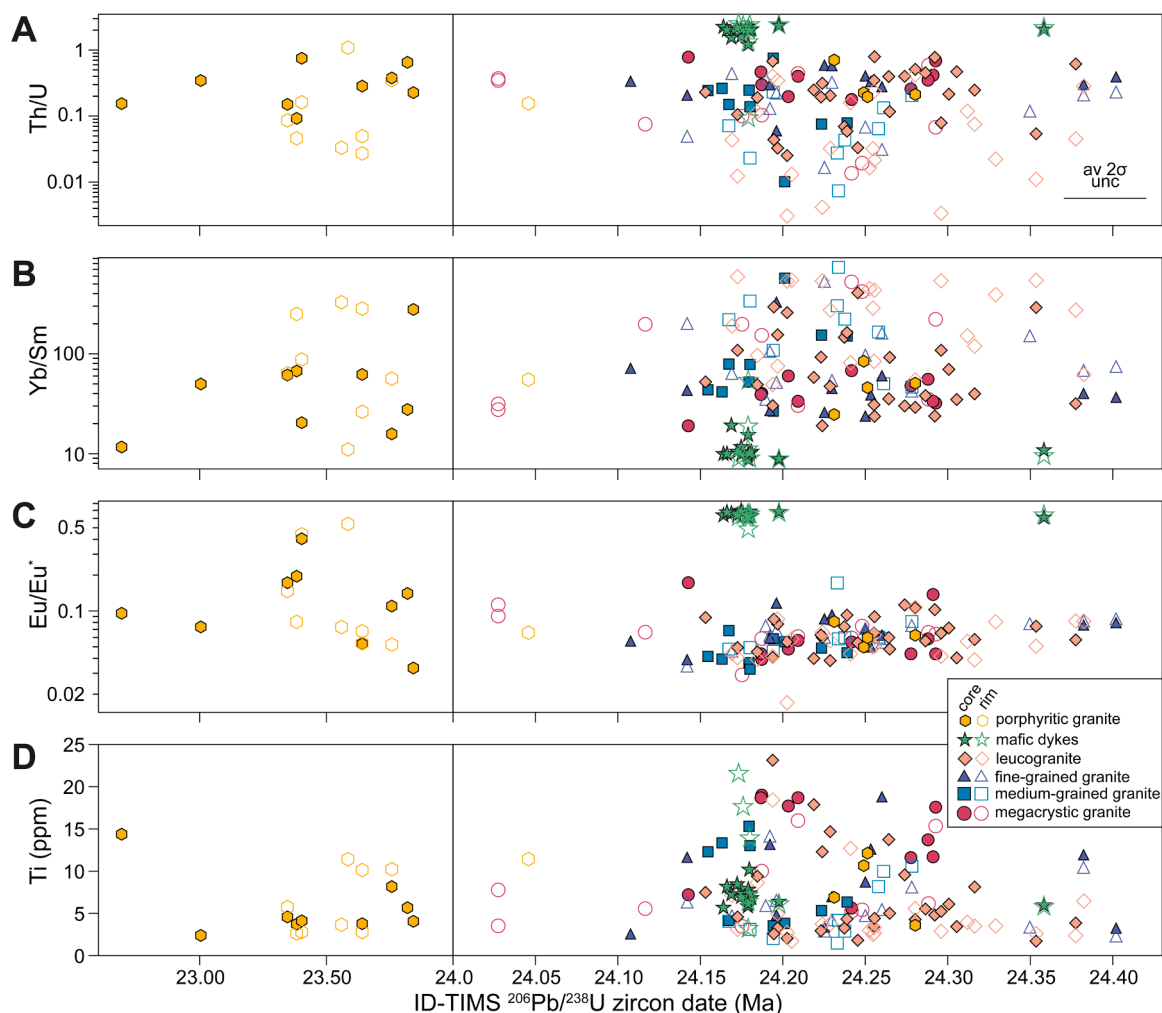


Fig. 6. Time-resolved (ID-TIMS) core-to-rim chemical variation of zircon crystals from different intrusions at San Rafael. Each point represents one in situ LA-ICPMS analysis (core or rim) acquired prior to whole-crystal ID-TIMS U–Pb dating, plotted against the respective whole crystal date. Note that some dates from Fig. 3 are not represented as the trace-element composition for these crystals could not be accurately determined due to presence of mineral or melt inclusions in the ablated portion of the zircon crystal. av 2σ unc – average 2σ uncertainty.

6. Discussion

6.1. Zircon record of magma evolution at San Rafael

The Oligocene zircon populations from all granites at San Rafael exhibit indistinguishable geochemical signatures (Figs. 5, 6 and S6–S7). The absence of correlation with magmatic unit and U–Pb dates suggests that zircon crystallized from a compositionally homogeneous magma body prior to emplacement at the deposit level (cf. Large et al., 2020; Virmond et al., 2024). The low Th/U ratios (0.15–0.30) in most cores and even lower ($\text{Th/U} < 0.15$) in rims, along with $\text{Eu/Eu}^* < 0.20$ in both cores and rims, indicate crystallization from a differentiating granitic melt that fractionated significant amounts of zircon and plagioclase (Fig. 5D). The progressive decrease in Ti contents from core to rim in most zircon crystals likely reflects predominant unidirectional cooling, without any extended temperature or magma chemistry reversals (Fig. 6D). In this context, the broad range of Ti contents observed in cores (~2–27 ppm; Fig. 4C), combined with small Ti differences between cores and rims within individual crystals (<5 ppm; Fig. 5D), suggest zircon nucleation over a wide range of temperatures, and subsequent crystal growth within a more restricted temperature span.

Zircon rare earth element (REE) signatures can be interpreted in light of their co-crystallizing mineral assemblage. Simple geochemical modeling demonstrates that the Sm content of zircon cores, increasing from ~15 ppm in the least evolved zircon (i.e., highest Ti content) to ~100 ppm, can be reproduced by early magmatic differentiation in the presence of zircon and apatite (and quartz, plagioclase, and K-feldspar in near eutectic proportion; see Fig. 5A, B and Supplementary Material), phases present in our samples (Harlaux et al., 2021a). Zircon Yb/Sm ratios <100 reflect this early differentiation stage (Fig. 5C). A shift towards lower MREE (e.g., Sm) and increasing HREE (e.g., Yb), recorded mostly in rims (Yb/Sm between 100 and 1000), is consistent with monazite joining the crystallizing assemblage at temperatures between 720 °C and 820 °C (at mid-crustal depths of 20 km and intrusion level of 2 km, respectively; Figs. 5A, B and S8). This temperature range also provides the depth-dependent minimum temperature estimate for melt generation in the crust. Monazite favors MREE over HREE in its structure (Stepanov et al., 2012), which leads to an increase in Yb while maintaining stable Sm contents in co-precipitating zircon (Fig. 5B). Crystallization of 0.02–0.05 vol. % monazite, consistent with its abundance in our samples (Harlaux et al., 2021a), reproduces the observed shift in Sm and Yb in zircon rims (Fig. 5A–C). Yb/Sm in zircon may thus serve as a general indicator of the relative timing of zircon and monazite crystallization, as well as the degree of differentiation of peraluminous melts. Our modelling also emphasizes that monazite significantly influences the REE budget of peraluminous melts that saturate zircon (Montel, 1993), highlighting the need to account for this effect before using zircon as a direct proxy for melt composition and oxidation state (e.g., Ce-based oxygen fugacity proxies; Gardiner et al., 2017; Roberts et al., 2024).

The two zircon populations identified in mafic dykes track the intrusion of mafic magmas into the granitic magma reservoir in the upper crust. The low Th/U-low Eu/Eu* population, texturally indistinguishable from zircon populations in the granites, was likely mechanically entrained during emplacement (Figs. 5D and S2). Contamination of mafic magmas by granitic material is consistent with mafic dykes showing highly variable chemical compositions (55–66 % SiO_2) and varying proportions of resorbed quartz grains (5–27 vol. %) and K-feldspar megacrysts (Harlaux et al. 2021a). In turn, prismatic zircon crystals from the mafic dyke sample SR23–9 (Figs. 2E and G, and S2) most likely crystallized at the deposit level upon intrusion of mafic melts into the granite. Rapid zircon crystallization upon saturation is supported by consistently high Th/U and Eu/Eu* ratios in these crystals (Fig. 5D) and overlapping high-precision dates (Fig. 4).

6.2. From high-precision U–Pb zircon dates to magmatic ages

High-precision ID-TIMS zircon U–Pb datasets often reveal age dispersions at the sample scale. While these age dispersions may keep record of underlying petrogenetic processes (Caricchi et al., 2014; Tavazzani et al., 2023; Nathwani et al., 2025), the youngest crystal in the population is typically interpreted to best approximate the age of final pluton crystallization or dyke/porphyry intrusion (von Quadt et al., 2011; Ratschbacher et al., 2018). The high-precision zircon dataset from San Rafael shows intrasample age dispersions of up to 1.7 Myr (Fig. 4), similar to what has been documented for granitic intrusions at Elba (Barboni et al., 2015) and Larderello (Farina et al., 2018), two other S-type systems studied recently at high precision. In this section we weigh the potential causes of this dispersion, considering magmatic and hydrothermal zircon crystallization, Pb loss, and cryptic inheritance of older zircon cores.

The coherent population between 24.402 ± 0.019 Ma and 24.006 ± 0.019 Ma most likely records the crystallization of peraluminous and mafic magmas throughout the crust until their semi-contemporaneous emplacement at shallow levels, as demonstrated by the range of comagmatic textures between granitic and mafic batches documented throughout the deposit (Figs. 2, 4, 7A; Harlaux et al. 2021a). These dates fall within the broad range of less precise, weighted mean in-situ zircon ages reported in Harlaux et al. (2021a), yet their higher precision now allows for constraining process durations in addition to their general timing. Twenty overlapping dates from the mafic dyke sample SR23–9 (Figs. 4 and 7B) are consistent with rapid crystallization upon emplacement in the upper crust. The remarkable consistency of these dates provides confidence that the dispersions observed in other samples genuinely reflect geological processes and are likely not influenced by analytical scatter. As this mafic dyke crystallized zircon upon intrusion into the megacrystic granite, its calculated weighted mean date of 24.177 ± 0.004 Ma provides a constraint on its emplacement age and establishes an age for the solidification of host megacrystic granite locally. If extrapolated, this age would also provide the timing of the pre-ore assembly and cooling of the complex in the upper crust (Fig. 7B). This interpretation aligns with the youngest zircon dates recorded in one sample of medium-grained and fine-grained granite each, and three out of four samples of leucogranite, all of which overlap with the emplacement age of the mafic dyke. However, the remaining pre-ore granite samples returned a few dates resolvable younger than the mafic dyke, potentially indicating either later emplacement or protracted, heterogeneous cooling of the complex until 23.9 Ma or beyond. The significantly younger dates from the same samples – up to ~400 kyr – could also indicate localized remelting of the granite by one or more younger magmatic pulses, with zircon crystallizing from this minimum melt (Asimus et al., 2024). This scenario, however, would require a substantial heat input at the emplacement level, for which there is no field evidence. Hydrothermal zircon growth, which provides another potential geological explanation for these younger dates, is likewise unsupported; these crystals not only display typical magmatic textures (Fig. S9) but also have Th/U ratios of 0.04–0.10 that are clearly higher than <<0.01 values characteristic of hydrothermal zircon (see Figs. 6 and S10; Hoskin and Schaltegger, 2003; Pettke et al., 2005).

Alternatively, the younger zircon dates from the pre-ore granites may reflect minor Pb loss, which would bias the interpretation of their true crystallization ages (Fig. 4). At U contents between 1000 and 19,000 ppm, San Rafael zircons have indeed been affected by non-negligible radiation damage, in spite of their young age. Radiation damage can promote the migration of radiogenic Pb, either between U-rich zones and adjacent zones within a crystal or out of the crystal entirely (Davis and Davis, 2018; Denyszyn et al., 2024). To minimize this effect, we applied chemical abrasion to all zircon crystals prior to high-precision analysis – a method shown to effectively remove domains affected by Pb loss while preserving those with undisturbed U–Pb isotope compositions (Mattinson, 2005; McKenna et al., 2024). Recognizing the

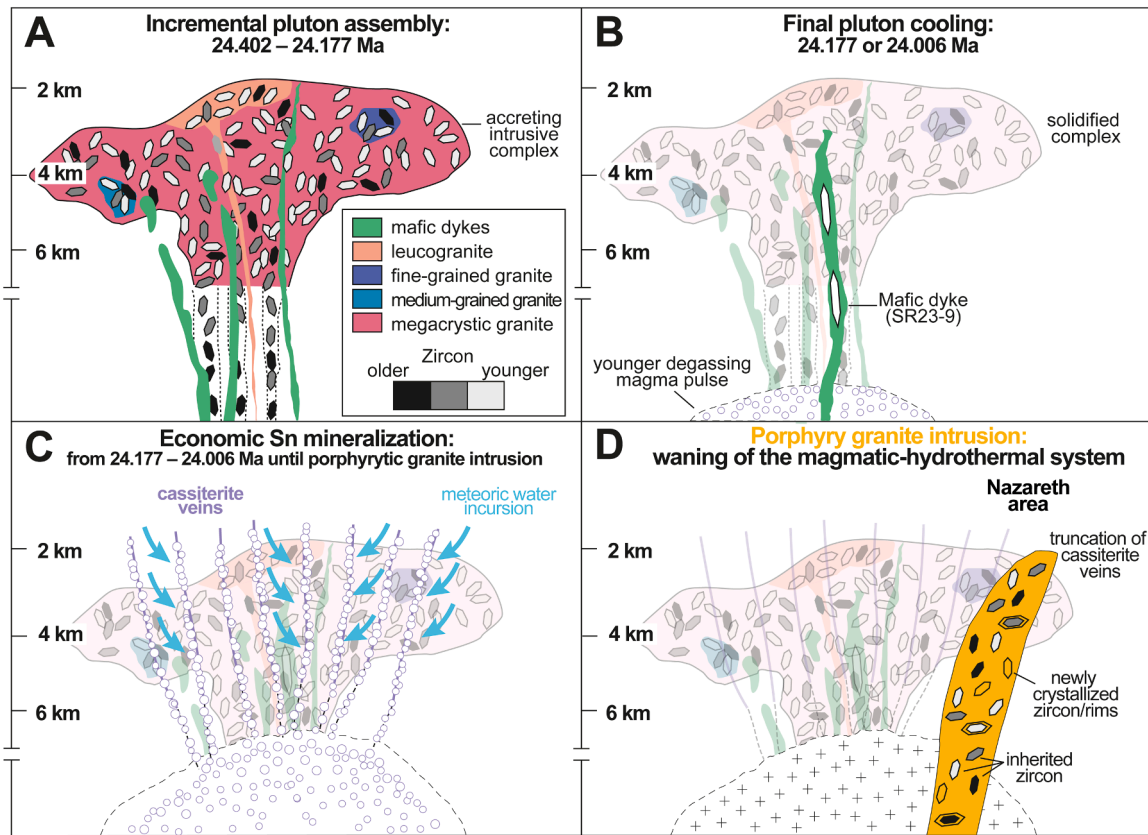


Fig. 7. Conceptual model for the evolution of the magmatic-hydrothermal Sn system at San Rafael. (A) Incremental assembly of the upper crustal intrusive complex by successive injections of silicic and mafic magma batches. Oligocene zircon population likely includes contributions from evolved melts in the lower to middle crust. (B) Final cooling of the granitic pluton is constrained by the weighted mean emplacement age of the mafic dyke (24.177 ± 0.004 Ma) or, alternatively, by the youngest zircon in the medium-grained granite (24.006 ± 0.019 Ma). (C) Hydrothermal cassiterite Sn mineralization occurred after the solidification of the San Rafael intrusive complex, possibly driven by degassing of a deeper magma body in the upper crust. Cassiterite formed through cooling of the magmatic fluid and mixing with meteoric waters at low pressures (e.g., Fekete et al., 2016; Korges et al., 2018; Harlaux et al., 2021b). (D) Intrusion of the porphyritic granite postdating Sn ore formation.

possibility of residual Pb loss not fully mitigated by chemical abrasion, we used zircon U and Th contents to estimate the accumulated α dose in the crystals' domains analyzed in situ prior to chemical abrasion (Fig. S5). We find that all analyses yield high α doses above $3 \times 10^{16} \alpha \text{ g}^{-1}$, and fourteen analyses – in crystals both from the younger as well as older group – exceed the empirical threshold of $6 \times 10^{17} \alpha \text{ g}^{-1}$ linked to Pb loss potential (McKanna et al., 2024). This finding underscores the overall higher susceptibility of San Rafael zircon to Pb loss. While the highest α doses suggest that some U-rich zones may have indeed suffered Pb loss, it is important to emphasize that the presence of high U, damaged zones does not imply that there still is unmitigated Pb loss in the dataset.

Zircon dates from the youngest porphyritic granite span 1.7 Myr, from ~ 24.37 Ma to 22.69 Ma (Fig. 4). Crosscutting relationships constrain the intrusion of the porphyritic granite to after the crystallization of the megacrystic host and precipitation of Sn ore, but before the onset of polymetallic mineralization (Figs. 2H–I, 3 and 7C–D; Harlaux et al. 2021a). The porphyritic granite stands out among the studied units, having 77 % of its zircon crystals showing evidence of inheritance, either as recycled, >30 Ma whole crystals or crystals containing >30 Ma cores overgrown by Oligocene rims (Fig. S11). By inference, much of the dispersion in whole-crystal high-precision dates may result from inclusion of small old cores (Barboni et al., 2015; Gaynor et al., 2023). A detailed inspection of CL textures of this population indicates that while the presence of a recycled core may be suspected in one crystal, all other crystals show continuous, magmatic zoning from core to rim (Fig. S9). In this context, the coherent date population between 24.37 Ma and 24.23

Ma is consistent with inheritance of whole crystals from the main magmatic pulse at San Rafael. The younger, more scattered dates could then represent mixed ages, which average a young magmatic rim age with varying amounts of inheritance undetected during cathodoluminescence imaging and in situ analyses. If we assume that the youngest date at 22.692 ± 0.020 Ma has the highest chance of being “truly magmatic”, this date could approximate the intrusion age of the porphyritic granite. While this interpretation is heavily reliant on a single zircon analysis, it aligns well with the crosscutting relationships. Moreover, this date closely matches $^{40}\text{Ar}/^{39}\text{Ar}$ ages of adularia associated with the late-stage polymetallic veins, which pinpoint the existence of a hydrothermal system active as late as between 22.72 and 22.29 Ma (Harlaux et al., 2023).

6.3. Timescales of Sn-ore-forming peraluminous magmatism

Thermal constraints require that granitic melt volumes similar to the estimates for the San Rafael complex ($\sim 72 \text{ km}^3$, based on a $4 \times 6 \text{ km}$ surface area and estimated $\sim 3 \text{ km}$ thickness; Fig. 1B–C) solidify at intrusion depths of $2\text{--}3 \text{ km}$ within just $\sim 50 \text{ kyr}$ (Kontak and Clark, 2002; Michel et al., 2008; Leuthold et al., 2012; Barboni et al., 2015). This implies that the $\sim 200 \text{ kyr}$ or more of dispersion in our zircon ages at the sample scale mostly reflects magmatic processes at deeper crustal levels — such as melt accumulation and transport — rather than upper crustal crystallization. Only the youngest zircon ages likely record the emplacement and final solidification of the San Rafael complex at the deposit level.

Own and literature crystallization ages of the main-stage cassiterite (Sn ore) and adularia associated with polymetallic ore are partially disconnected from the main population of magmatic zircon ages (Fig. 4). Compared to ID-TIMS zircon ages, uncertainties on cassiterite and adularia ages are an order of magnitude larger (1–4 % and ~0.1 % at 2 σ level), which challenges a direct discrimination of absolute time relationships between the magmatic and hydrothermal stages (cf. Harlaux et al., 2023). Nonetheless, we observe that while most cassiterite ages broadly overlap with the intrusive complex's crystallization age at 24.177 ± 0.004 Ma, one coarse-grained cassiterite sample is ~200 kyr younger, and adularia ages are up to ~1.9 Myr younger (Fig. 4; Harlaux et al., 2023). Considering the rapid cooling and crystallization expected at 2–3 km depth (Kontak and Clark, 2002; Barboni et al., 2015), we deduce that it is unlikely that cassiterite and polymetallic ore are genetically related to cooling of the host intrusive complex. Instead, they are more likely linked to a younger degassing magmatic pulse concealed at depth (Fig. 7C; Harlaux et al., 2021a). Alternatively, the spread of cassiterite and adularia ages may reflect multiple events of Sn and (telescoped?) polymetallic mineralization, each tied to separate magmatic pulses. The youngest porphyritic granite could be an expression of one such later pulse; this unit shows a strong alteration overprint (Harlaux et al., 2021a) and its inferred intrusion age at 22.692 ± 0.020 Ma or younger closely aligns with adularia ages marking polymetallic mineralization stages III and IV (~22.72–22.43 Ma; Figs. 3 and 7D). Despite these differences in timing, zircon geochemistry reveals that both the pre-ore and porphyritic granites formed under similar magmatic conditions. This indicates that, even if a particular magmatic pulse were more likely to generate Sn mineralization, such a signal is not resolved in the zircon record. Mafic melts have instead been proposed to play a key role in Sn ore formation by triggering hydrothermal fluid release when intruding into the unconsolidated granite (Harlaux et al., 2021a). Our zircon petrochronology of a mafic dyke confirms the coexistence of mafic and granitic magmas in the upper crustal reservoir, though it provides no further insight into the role of mafic melts in ore formation. As a general consideration, for mafic magmas to be regarded as significant contributors of heat, fluids, or—particularly—metals on a deposit scale, it would need to be demonstrated that they occur in volumes much larger than the scattered enclaves and dykes currently observed.

6.4. Limits to sizes of Sn magmatic-hydrothermal systems

The formation of magmatic-hydrothermal deposits involves the transfer and focusing of metals from outgassing magma reservoirs via hydrothermal fluids to ore precipitation sites in the upper crust (<5 km; e.g., Hedenquist and Lowenstern, 1994). Both Sn deposits associated with S-type granites and porphyry copper deposits follow this underlying genetic mechanism, despite differences in mineralization styles (e.g., veins, disseminations, skarn) that reflect the shallow physico-chemical conditions where metals precipitate. However, despite this common origin, these two deposit types differ dramatically in scale; a simple comparison shows that San Rafael – being the world's largest S-type magmatic-hydrothermal Sn deposit – contains roughly a hundred times less metal than some of the largest porphyry deposits, such as the “behemothian” Chuquicamata that hosts ~95 Mt of Cu (Virmond et al., 2024).

This stark difference in metal endowment might arise from vastly contrasting volumes of parental magmas that outgas in a focused manner to contribute fluid and metals for ore formation in S-type Sn versus porphyry Cu systems – a concept referred to as “effective magma volume” by Chelle-Michou and Rottier (2021). To estimate the required magma volumes, we assumed that half the mass of metal from melt is precipitated in ore (50 % precipitation efficiency), an initial concentration of 15 ppm in peraluminous melt for San Rafael (based on the mean Sn concentration in unaltered samples from the host megacrystic complex; Harlaux et al. 2021a), and 30 ppm of Cu in calc-alkaline melt

for porphyry copper systems (see compilation by Virmond et al., 2024). Our estimate for San Rafael presupposes a tenfold enrichment of Sn in melt relative to average continental crust (Rudnick and Gao, 2003), consistent with lower-end values reported from melt inclusions in the Bolivian Tin Belt (Dietrich et al., 2000). With these parameters, producing 1.5 Mt of Sn at San Rafael would require 77 km³ of effective magma (Fig. S12) – about the same volume as the upper crustal intrusive complex alone, which has been shown unlikely to be the source of Sn. This mass balance underscores a key point: that the magmatic system as a whole contains more than enough Sn to support a much larger endowment, indicating that magmatic Sn availability is not the limiting factor in the formation of Sn deposits. The 1.5 Mt of Sn is indeed best viewed as a minimum, especially in light of recent exploration suggesting that San Rafael may be part of a larger Sn resource associated with an exceptionally voluminous magmatic-hydrothermal system in the Macusani Structural Zone. The large Santo Domingo Sn(-Cu) deposit, located about 22 km from San Rafael, shares striking similarities in system architecture, age (Clark et al., 1983), and structural patterns, raising the possibility that the two might be upper-crustal expressions of a single, outgassing transcrustal system derived from a common source. Nonetheless, even if combined magma volumes could exceed 200 km³, the total Sn endowment from San Rafael and Santo Domingo would, even in most generous estimates, likely remain below 10 Mt – a small fraction of the Cu content hosted by the largest porphyry systems.

From this comparison, we posit that – unless there are unrecognized differences in magma cooling rates that govern fluid production (Chelle-Michou et al., 2017a) – the contrasting metal endowments in S-type Sn versus porphyry Cu systems may reflect different petrogenetic modes of magma generation in (post-)collisional versus arc settings. While our primary emphasis is on magmatic processes, we recognize that variations in the initial metal content of parental magmas, metal partitioning between melt and fluid, and differences in metal complexation may also contribute (e.g., Zajacz et al., 2008; Tattitch and Blundy, 2017; Schmidt et al., 2020). In porphyry systems, the largest copper deposits have been directly linked to large effective magma volumes (>1000 km³) and multi-million-long durations of magmatic-hydrothermal activity (Chelle-Michou et al., 2017a; Chiaradia and Caricchi, 2017; Chelle-Michou and Rottier, 2021; Virmond et al., 2024). In magmatic arcs, such volumes can be supplied continuously to upper crustal reservoirs over extended timescales, partly due to long-lived deep crustal hot zones, which promote efficient heat transfer from the mantle and thermally precondition the overlying crust (e.g., Annen, 2009; Gelman et al., 2013; Karakas et al., 2017; Schöpa et al., 2017). These arc magmas also tend to have higher water contents, sometimes exceeding 5 wt. % (Loucks et al., 2014; Nathwani et al., 2024), which enhances the transfer of metals into hydrothermal fluids (Lamy-Chappuis et al., 2020; Rezeau and Jagoutz, 2020; Gruzdeva et al., 2024). In contrast, S-type systems generally experience more limited and episodic melt supply (Moyen et al., 2021; Pichavant, 2024), which likely hinders the accretion of comparably large effective magma volumes over similar timescales. This difference reflects a fundamental distinction in source dynamics: while mantle-derived systems benefit from a more extensive and potentially replenishable melt source – sustained by convection – crustal sources are spatially constrained and can only yield a finite volume of melt under favorable conditions. Crustal melting depends critically on the presence of fertile lithologies and the availability of water to facilitate melting. Together, these factors constrain the generation of large effective magma volumes in S-type systems and constitute a key process-level bottleneck that limits the maximum size of associated Sn magmatic-hydrothermal systems.

7. Conclusions

We applied zircon petrochronology to a suite of intrusions at San Rafael, the world's largest and highest-grade magmatic-hydrothermal Sn deposit (>1.5 Mt of Sn at 3.7 %). Our high-precision U-Pb zircon ages

from pre-ore granites and coeval mafic units indicate that the full evolution of the crustal magmatic system occurred within a timespan of 200–400 kyr – from 24.402 ± 0.019 Ma to the final emplacement and crystallization of the shallow-level intrusive complex (24.177 ± 0.004 Ma or younger). Zircon compositions (e.g., Ti and REE contents) are indistinguishable across all granites, suggesting stable magmatic conditions throughout the system's lifespan. Ages of cassiterite ore and hydrothermal adularia – some of which postdate the intrusive complex emplacement by >200 kyr – indicate that Sn ore most likely did not derive from fluids exsolved by these granites but instead from one or more younger magma pulses not currently exposed at deposit depth. Our analysis of zircon age dispersions underscores the need for improved understanding of Pb loss mechanisms and cryptic inheritance in zircon, in order to further enhance confidence in age interpretations and fully exploit the resolving power of high-precision U–Pb datasets. At San Rafael, exceptionally high U contents in zircon (up to >10,000 ppm) have led to significant radiation damage, increasing susceptibility to Pb loss even at a relatively young age. The large age scatter (>1.7 Myr) observed in the youngest porphyritic granite is consistent with recycling of small inherited zircon cores into whole-crystal ID-TIMS analyses, potentially biasing inferred timescales. As a strategy to evade this effect, microsampling of outermost magmatic rims holds promise as a way of isolating the pristine magmatic crystallization signal in systems with ample inheritance (Markovic et al., 2024). More broadly, our study presents the concept that episodic melt supply in post-collisional S-type settings and limited water availability represent fundamental constraints on the size of Sn magmatic-hydrothermal systems. Looking ahead, integrating high-precision geochronology with thermal and fluid flow modelling will be crucial for placing quantitative constraints on the processes that govern the size and evolution of tin systems.

CRediT authorship contribution statement

Sava Markovic: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Dawid Szymanowski:** Writing – review & editing, Validation, Methodology, Investigation. **Lorenzo Tavazzani:** Writing – review & editing, Validation, Methodology, Formal analysis. **Lisard Torró:** Writing – review & editing, Validation, Investigation. **Kalin Kouzmanov:** Writing – review & editing, Resources. **Miroslav Kalinaj:** Writing – review & editing, Resources. **Cyril Chelle-Michou:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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Data availability

The data is available as an electronic supplement

References

- Annen, C., 2009. From plutons to magma chambers: thermal constraints on the accumulation of eruptible silicic magma in the upper crust. *Earth. Planet. Sci. Lett.* 284, 409–416. <https://doi.org/10.1016/j.epsl.2009.05.006>.
- Asimus, J.L., Daczko, N.R., Gazi, J.A., Ezad, I.S., Belousov, I., Rodemann, T., Halpin, J.A., Piazzolo, S., 2024. Experimental replacement of zircon by melt-mediated coupled dissolution–precipitation causes dispersion in U–Pb ages. *J. Metamorph. Geol.* 42, 1179–1195. <https://doi.org/10.1111/jmg.12795>.
- Barboni, M., Annen, C., Schoene, B., 2015. Evaluating the construction and evolution of upper crustal magma reservoirs with coupled U/Pb zircon geochronology and thermal modeling: a case study from the Mt. Capanne pluton (Elba, Italy). *Earth. Planet. Sci. Lett.* 432, 436–448. <https://doi.org/10.1016/j.epsl.2015.09.043>.
- Bermejo, D., Ortega, L., Sánchez, S.B., Tavazzani, L., Castañeiras, P., Chelle-Michou, C., Crespo, E., Alves, K.D.S., Gómez-Barreiro, J., 2025. Late variscan tectonic orogenic collapse as a trigger of Sn–W mineralizing systems. U–Pb ore geochronology across the Martinamor gneissic dome (Salamanca, Spain). *Ore Geol. Rev.*, 106762 <https://doi.org/10.1016/j.oregeorev.2025.106762>.
- Caricchi, L., Simpson, G., Schaltegger, U., 2014. Zircons reveal magma fluxes in the Earth's crust. *Nature* 511, 457–460. <https://doi.org/10.1038/nature13532>.
- Carlier, G., Lorand, J.P., Liégeois, J.P., Fornari, M., 2005. Potassic–ultrapotassic mafic rocks delineate two-lithospheric mantle blocks beneath the southern Peruvian Altiplano. *Geology* 33, 601–604. <https://doi.org/10.1130/G21643.1>.
- Carr, P.A., Moreira, E., Neymark, L., Norman, M.D., Mercadier, J., 2023. A LA–ICP–MS comparison of reference materials used in cassiterite U–Pb geochronology. *Geostand. Geoanal. Res.* 47, 67–87. <https://doi.org/10.1111/ggr.12469>.
- Černý, P., Blevin, P.L., Cuney, M., London, D., 2005. Granite-related ore deposits. In: Hedenquist, J.W., Thompson, J.F.H., Goldfarb, R.J., Richards, J.P. (Eds.), *One Hundredth Anniversary Volume. Society of Economic Geologists*, pp. 337–370. <https://doi.org/10.5382/AV100.12>.
- Chelle-Michou, C., Rottier, B., Caricchi, L., Simpson, G., 2017a. Tempo of magma degassing and the genesis of porphyry copper deposits. *Sci. Rep.* 7, 40566. <https://doi.org/10.1038/srep40566>.
- Chelle-Michou, C., Rottier, B., 2021. Transcrustal magmatic controls on the size of porphyry Cu systems: state of knowledge and open questions. *Society of Economic Geologists Special Publication* 24, 87–100. <https://doi.org/10.5382/SP.24.06>.
- Chiaradia, M., Caricchi, L., 2017. Stochastic modelling of deep magmatic controls on porphyry copper deposit endowment. *Sci. Rep.* 7, 44523. <https://doi.org/10.1038/srep44523>.
- Clark, A.H., Farrar, E., Kontak, D.J., Langridge, R.J., Arenas, M.J., France, L.J., McBride, S.L., Woodman, P.L., Wasteneys, H.A., Sandeman, H.A., Archibald, D.A., 1990. Geologic and geochronological constraints on the metallogenic evolution of the Andes of southeastern Peru. *Econ. Geol.* 85, 1520–1583. <https://doi.org/10.2113/gsecongeo.85.7.1520>.
- M.J. Clark, A.H., Palma, V.V., Archibald, D.A., Farrar, E., Arenas, F., Robertson, R.C.R., 1983. Occurrence and age of tin mineralization in the Cordillera Oriental, southern Peru. *Econ. Geol.* 78, 514–520. <https://doi.org/10.2113/gsecongeo.78.3.514>.
- Davis, W.J., Davis, D.W., 2018. Alpha recoil loss of Pb from baddeleyite evaluated by high-resolution ion microprobe (SHRIMP II) depth profiling and numerical modeling: implications for the interpretation of U–Pb ages in small baddeleyite crystals. In: Reiners, P.W., Gautheron, D. (Eds.), *Microstructural Geochronology: Planetary Records Down to Atom Scale*, pp. 247–259. <https://doi.org/10.1002/9781119227250.ch11>.
- Denyszyn, S., Davis, D.W., Fougereuse, D., 2024. Nanoscale heterogeneity of U and Pb in baddeleyite—implications for nanogeochronology and ^{238}U series alpha recoil effects. *Geochronology* 6, 607–619. <https://doi.org/10.5194/gchron-6-607-2024>.
- Dietrich, A., Lehmann, B., Wallianos, A., 2000. Bulk rock and melt inclusion geochemistry of Bolivian tin porphyry systems. *Econ. Geol.* 95, 313–326. <https://doi.org/10.2113/gsecongeo.95.2.313>.
- Farina, F., Dini, A., Davies, J.H.F.L., Ovtcharova, M., Greber, N.D., Bouvier, A.S., Baumgartner, L., Ulianov, A., Schaltegger, U., 2018. Zircon petrochronology reveals the timescale and mechanism of anatectic magma formation. *Earth. Planet. Sci. Lett.* 495, 213–223. <https://doi.org/10.1016/j.epsl.2018.05.021>.
- Fekete, S., Weis, P., Driesner, T., Bouvier, A.S., Baumgartner, L., Heinrich, C.A., 2016. Contrasting hydrological processes of meteoric water incursion during magmatic–hydrothermal ore deposition: an oxygen isotope study by ion microprobe. *Earth. Planet. Sci. Lett.* 451, 263–271. <https://doi.org/10.1016/j.epsl.2016.07.009>.
- Gardiner, N.J., Hawkesworth, C.J., Robb, L.J., Whitehouse, M.J., Roberts, N.M.W., Kirkland, C.L., Evans, N.J., 2017. Contrasting granite metallogeny through the zircon record: a case study from Myanmar. *Sci. Rep.* 7, 748. <https://doi.org/10.1038/s41598-017-00832-2>.

- Gaynor, S.P., Smith, T.M., Schaltegger, U., 2023. Tracing magmatic genesis and evolution through single zircon crystals from successive supereruptions from the Socorro Caldera Complex, USA. *Earth. Planet. Sci. Lett.* 616, 118236. <https://doi.org/10.1016/j.epsl.2023.118236>.
- Gelman, S.E., Gutiérrez, F.J., Bachmann, O., 2013. On the longevity of large upper crustal silicic magma reservoirs. *Geology*, 41, 759–762. <https://doi.org/10.1130/G34241.1>.
- Gemmrich, L., Torró, L., Melgarejo, J.C., Laurent, O., Vallance, J., Chelle-Michou, C., Sempere, T.P.A., 2021. Trace element composition and U–Pb ages of cassiterite from the Bolivian tin belt. *Miner. Deposita* 56, 1491–1520. <https://doi.org/10.1007/s00126-020-01030-3>.
- Gialli, S., Chauvet, A., Kouzmanov, K., Dini, A., Spikings, R., Kalinaj, M., Fontboté, L., 2017. Structural Control of the World-Class Sn district of San Rafael, Peru, 2017. *Ore Deposits of China and Beyond*, Society of Economic Geologists, Beijing, China, p. 206. September 17–20.
- Gruzdeva, Y., Weis, P., Andersen, C., 2024. Timing of volatile degassing from hydrous upper-crustal magma reservoirs with implications for porphyry copper deposits. *J. Geophys. Res. Solid. Earth*, 129, e2023JB028433. <https://doi.org/10.1029/2023JB028433>.
- Harlaux, M., Kontak, D.J., Clark, A.H., Kouzmanov, K., Holm-Denoma, C.S., Gialli, S., Laurent, O., Spikings, R., Chauvet, A., Dini, A., Kalinaj, M., Fontboté, L., 2023. Depositing >1.5 Mt of tin within <1 my. of initial granitic intrusion in the San Rafael tin (copper) deposit, southeastern Peru. *Econ. Geol.* 118, 1371–1396. <https://doi.org/10.5382/econgeo.5021>.
- Harlaux, M., Kouzmanov, K., Gialli, S., Clark, A.H., Laurent, O., Corthay, G., Flores, E.P., Dini, A., Chauvet, A., Ulianov, A., Chiaradia, M., Menzies, A., Durand, G.V., Kalinaj, M., Fontboté, L., 2021a. The upper oligocene San Rafael intrusive complex (Eastern Cordillera, southeast Peru), host of the largest-known high-grade tin deposit. *Lithos*, 400, 106409. <https://doi.org/10.1016/j.lithos.2021.106409>.
- Harlaux, M., Kouzmanov, K., Gialli, S., Marger, K., Bouvier, A.S., Baumgartner, L.P., Rielli, A., Dini, A., Chauvet, A., Kalinaj, M., Fontboté, L., 2021b. Fluid mixing as primary trigger for cassiterite deposition: evidence from in situ 8180–811B analysis of tourmaline from the world-class San Rafael tin (copper) deposit. *Peru. Earth Planet. Sci. Lett.* 563, 116889. <https://doi.org/10.1016/j.epsl.2021.116889>.
- Hedenquist, J.W., Lowenstern, J.L., 1994. The role of magmas in the formation of hydrothermal ore deposits. *Nature* 370, 519–527. <https://doi.org/10.1038/370519a0>.
- Hoskin, P.W.O., Schaltegger, U., 2003. The composition of zircon and igneous and metamorphic petrogenesis. *Rev. Mineral. Geochem.* 53, 27–62. <https://doi.org/10.2113/0530027>.
- Heinrich, C.A., 1990. The chemistry of hydrothermal tin (tungsten) ore deposition. *Econ. Geol.* 85, 457–481. <https://doi.org/10.2113/gsecongeo.85.3.457>.
- Karakas, O., Degruyter, W., Bachmann, O., Dufek, J., 2017. Lifetime and size of shallow magma bodies controlled by crustal-scale magmatism. *Nat. Geosci.* 10, 446–450. <https://doi.org/10.1038/ngeo2959>.
- Keppeler, H., Audétat, A., 2025. The redox dependence of the fluid/melt partitioning of tin. *Geochim. Cosmochim. Acta* 394, 182–193. <https://doi.org/10.1016/j.gca.2025.02.007>.
- Kontak, D.J., Clark, A.H., 2002. Genesis of the giant, bonanza San Rafael Lode tin deposit, Peru: origin and significance of pervasive alteration. *Econ. Geol.* 97, 1741–1777. <https://doi.org/10.2113/97.8.1741>.
- Korges, M., Weis, P., Lüders, V., Laurent, O., 2018. Depressurization and boiling of a single magmatic fluid as a mechanism for tin-tungsten deposit formation. *Geology*, 46, 75–78. <https://doi.org/10.1130/G39601.1>.
- Kuiper, K.F., Deino, A., Hilgen, F.J., Krijgsman, W., Renne, P.R., Wijbrans, J.R., 2008. Synchronizing rock clocks of Earth history. *Science* (1979) 320, 500–504. <https://doi.org/10.1126/science.1154339>.
- Lamy-Chappuis, B., Heinrich, C.A., Driesner, T., Weis, P., 2020. Mechanisms and patterns of magmatic fluid transport in cooling hydrous intrusions. *Earth. Planet. Sci. Lett.* 535, 11611. <https://doi.org/10.1016/j.epsl.2020.116111>.
- Large, S.J.E., Wotzlaw, J.F., Guillong, M., von Quadt, A., Heinrich, C.A., 2020. Resolving the timescales of magmatic and hydrothermal processes associated with porphyry deposit formation using zircon U–Pb petrochronology. *Geochronology* 2, 209–230. <https://doi.org/10.5194/gchron-2-209-2020>.
- Laubacher, G., 1978. *Estudio Geológico De La Region Norte Del Lago Titicaca*. Instituto Geología y Minería, Peru.
- Lehmann, B., Ishihara, S., Michel, H., Miller, J., Rapela, C.W., Sánchez, A., Tisl, M., Winkelmann, L., 1990. The Bolivian tin province and regional tin distribution in the Central Andes: a reassessment. *Econ. Geol.* 85 (5), 1044–1058. <https://doi.org/10.2113/gsecongeo.85.5.1044>.
- Lehmann, B., 2021. Formation of tin ore deposits: a reassessment. *Lithos*, 402–403, 105756. <https://doi.org/10.1016/j.lithos.2020.105756>.
- Leuthold, J., Müntener, O., Baumgartner, L.P., Putlitz, B., Ovtcharova, M., Schaltegger, U., 2012. Time resolved construction of a bimodal laccolith (Torres del Paine, Patagonia). *Earth. Planet. Sci. Lett.* 325, 85–92. <https://doi.org/10.1016/j.epsl.2012.01.032>.
- Loucks, R.R., 2014. Distinctive composition of copper-ore-forming arc magmas. *Aust. J. Earth Sci.* 61, 6–16. <https://doi.org/10.1080/08120099.2013.865676>.
- Mamani, M., Navarro, P., Carlotto, V., Acosta, H., Rodríguez, J., Jaimes, F., Cereceda, C., 2010. Arcos magmáticos meso-cenozoicos del Perú. *Soc. Geol. Perú, XV Congr. Peruano Geol., Cusco. Resúmenes Ext* 563–566.
- Mattinson, J.M., 2005. Zircon U–Pb chemical abrasion (“CA-TIMS”) method: combined annealing and multi-step partial dissolution analysis for improved precision and accuracy of zircon ages. *Chem. Geol.* 220, 47–66. <https://doi.org/10.1016/j.chemgeo.2005.03.011>.
- McKenna, A.J., Schoene, B., Szymanowski, D., 2024. Geochronological and geochemical effects of zircon chemical abrasion: insights from single-crystal stepwise dissolution experiments. *Geochronology* 6, 1–20. <https://doi.org/10.5194/gchron-6-1-2024>.
- Michel, J., Baumgartner, L., Putlitz, B., Schaltegger, U., Ovtcharova, M., 2008. Incremental growth of the Patagonian Torres del Paine laccolith over 90 k.y. *Geology*, 36, 459–462. <https://doi.org/10.1130/G24546a.1>.
- Minsur, S.A., 2023. Annual Report. Available at www.minsur.com. last accessed 7 April 2025.
- Montel, J.M., 1993. A model for Monazite/melt equilibrium and application to the generation of granitic magmas. *Chem. Geol.* 110, 127–146. [https://doi.org/10.1016/0009-2541\(93\)90250-M](https://doi.org/10.1016/0009-2541(93)90250-M).
- Moyen, J.F., Janousek, V., Laurent, O., Bachmann, O., Jacob, J.B., Farina, F., Fiannacca, P., Villaros, A., 2021. Crustal melting vs. fractionation of basaltic magmas: part 1, granites and paradigms. *Lithos*, 402–403, 106291. <https://doi.org/10.1016/j.lithos.2021.106291>.
- Nathwani, C.L., Szymanowski, D., Tavazzani, L., Markovic, S., Virmond, A.L., Chelle-Michou, C., 2025. Controls on zircon age distributions in volcanic, porphyry, and plutonic rocks. *Geochronology* 7, 15–33. <https://doi.org/10.5194/gchron-7-15-2025>.
- Nathwani, C., Blundy, J., Large, S.J.E., Wilkinson, J.J., Buret, Y., Loader, M.A., Tavazzani, L., Chelle-Michou, C., 2024. A zircon case for super-wet arc magmas. *Nat. Commun.* 15, 8982. <https://doi.org/10.1038/s41467-024-52786-5>.
- Patterson, D.J., Ohmoto, H., Solomon, M., 1981. Geologic setting and genesis of cassiterite-sulfide mineralization at Renison Bell, Western Tasmania. *Econ. Geol.* 76, 393–438. <https://doi.org/10.2113/gsecongeo.76.2.393>.
- Perez, N.D., Horton, B.K., Carlotto, V., 2016. Structural inheritance and selective reactivation in the central Andes: cenozoic deformation guided by pre-Andean structures in southern Peru. *Tectonophysics*, 671, 264–280. <https://doi.org/10.1016/j.tecto.2015.12.031>.
- Pettke, T., Audétat, A., Schaltegger, U., Heinrich, C.A., 2005. Magmatic-to-hydrothermal crystallization in the W–Sn mineralized mole granite (NSW, Australia) – Part II: evolving zircon and thorite trace element chemistry. *Chem. Geol.* 220, 191–213. <https://doi.org/10.1016/j.chemgeo.2005.02.017>.
- Pichavant, M., Villaros, A., Michaud, J.A.S., Scaillet, B., 2024. Granite magmatism and mantle filiation. *Eur. J. Mineral.* 36, 225–246. <https://doi.org/10.5194/ejm-36-225-2024>.
- Ratschbacher, B.C., Keller, C.B., Schoene, B., Paterson, S.R., Anderson, J.L., Okaya, D., Putirka, K., Lippoldt, R., 2018. A new workflow to assess emplacement duration and melt residence time of compositionally diverse magmas emplaced in a sub-volcanic reservoir. *J. Petrol.* 59, 1787–1809. <https://doi.org/10.1093/petrology/egy079>.
- Rezeau, H., Jagoutz, O., 2020. The importance of H₂O in arc magmas for the formation of porphyry Cu deposits. *Ore Geol. Rev.* 126, 103744.
- Roberts, N.M.W., Yakymchuk, C., Spencer, C.J., Keller, C.B., Tapster, S.R., 2024. Revisiting the discrimination and distribution of S-type granites from zircon trace element composition. *Earth. Planet. Sci. Lett.* 633, 118638. <https://doi.org/10.1016/j.epsl.2024.118638>.
- Rudnick, R.L., Gao, S., 2003. Composition of the continental crust. In: Rudnick, R.L. (Ed.), *The Crust. Treatise on Geochemistry, The Crust. Treatise on Geochemistry*, 3. Elsevier, Amsterdam, pp. 1–64. <https://doi.org/10.1016/B0-08-043751-6/03016-4>.
- Sandeman, H.A., Clark, A.H., Farrar, E., 1995. An integrated tectono-magmatic model for the evolution of the southern Peruvian Andes (13–20°S) since 55 ma. *Int. Geol. Rev.* 37, 1039–1073. <https://doi.org/10.1080/00206819509465439>.
- Sandeman, H.A., Clark, A.H., Farrar, E., Pauca, G.A., 1997. Lithostratigraphy, petrology and ⁴⁰Ar/³⁹Ar geochronology of the Crucero Supergroup, Puno Department, SE Peru. *J. S. Am. Earth Sci.* 10, 223–245. [https://doi.org/10.1016/S0895-9811\(97\)00023-0](https://doi.org/10.1016/S0895-9811(97)00023-0).
- Schöpa, A., Annen, C., Dilles, J.H., Sparks, R.S.J., Blundy, J.D., 2017. Magma emplacement rates and porphyry copper deposits: thermal modeling of the Yerington Batholith, Nevada. *Econ. Geol.* 112, 1653–1672. <https://doi.org/10.5382/econgeo.2017.4525>.
- Schmidt, C., Romer, R.L., Wohlgemuth-Ueberwasser, C.C., Appelt, O., 2020. Partitioning of Sn and W between granitic melt and aqueous fluid. *Ore Geol. Rev.* 117, 103263. <https://doi.org/10.1016/j.oregeorev.2019.103263>.
- Stepanov, A.S., Hermann, J., Rubatto, D., Rapp, R.P., 2012. Experimental study of monazite/melt partitioning with implications for the REE, Th and U geochemistry of crustal rocks. *Chem. Geol.* 300, 200–220. <https://doi.org/10.1016/j.chemgeo.2012.01.007>.
- Tattitch, B.C., Blundy, J.D., 2017. Cu–Mo partitioning between felsic melts and saline-aqueous fluids as a function of X_{Na}Cleq, fO₂, and fS₂. *American Mineralogist* 102 (10), 1987–2006. <https://doi.org/10.2138/am-2017-5998>.
- Tavazzani, L., Wotzlaw, J.F., Economos, R., Sinigoi, S., Demarchi, G., Szymanowski, D., Laurent, O., Bachmann, O., Chelle-Michou, C., 2023. High-precision zircon age spectra record the dynamics and evolution of large open-system silicic magma reservoirs. *Earth. Planet. Sci. Lett.* 623. <https://doi.org/10.1016/j.epsl.2023.118432>.
- Torró, L., Harlaux, M., Castro-Morante, A., Vallance, J., Tavazzani, L., Bouvier, A.S., Bovay, T., Chelle-Michou, C., Sempere, T., Melgarejo, J.C., 2024. Tin mineralization in the triassic Chacaltaya District (Cordillera Real, Bolivia) traced by in situ chemical and $\delta^{18}\text{O}$ – $\delta^{31}\text{P}$ compositions of tourmaline. *Econ. Geol.* 119, 471–500. <https://doi.org/10.5382/econgeo.5051>.
- Virmond, A.L., Wotzlaw, J.-F., Rojas-Arancibia, R., Selby, D., Chelle-Michou, C., 2024. Multi-million-year magmatic and hydrothermal activity is key to the formation of

- supergiant to behemothian porphyry copper deposits. *Contrib. Mineral. Petrol.* 179. <https://doi.org/10.1007/s00410-024-02167-4>.
- von Quadt, A., Erni, M., Martinek, K., Moll, M., Peytcheva, I., Heinrich, C.A., 2011. Zircon crystallization and the lifetimes of ore-forming magmatic-hydrothermal systems. *Geology*. 39, 731–734. <https://doi.org/10.1130/G31966.1>.
- Zajacz, Z., Halter, W.E., Pettke, T., Guillong, M., 2008. Determination of fluid/melt partition coefficients by LA-ICPMS analysis of co-existing fluid and silicate melt inclusions: controls on element partitioning. *Geochim. Cosmochim. Acta* 72 (8), 2169–2197. <https://doi.org/10.1016/j.gca.2008.01.034>.